

SUMMARY

Final-year CS student at FAST-NUCES with 1.5 years of backend development experience in Python, Django, FastAPI, and PostgreSQL. Built and deployed REST APIs serving 10,000+ daily active users. Strong grasp of system design, containerization, and CI/CD. Seeking a backend engineering role.

EDUCATION

BS Computer Science — FAST-NUCES, Karachi

2020 – 2024 | GPA: 3.7 / 4.0 | Dean's List 2022, 2023

- Relevant Coursework: Data Structures, Algorithms, Database Systems, Operating Systems, Software Engineering, Computer Networks
- Teaching Assistant for Data Structures course (Spring 2023) — graded assignments for 120 students

SKILLS

Languages: Python, Java, C++, SQL, Bash

Frameworks: Django REST Framework, FastAPI, Flask, Spring Boot, Celery

Databases: PostgreSQL, MySQL, Redis, MongoDB

Tools: Docker, Git, Linux, Nginx, RabbitMQ, GitHub Actions

Other: REST APIs, System Design, Unit Testing, CI/CD, JWT, OAuth2

EXPERIENCE

Backend Developer Intern — Walee Technologies, Karachi

June 2023 – December 2023

- Developed 12 REST API endpoints using Django REST Framework for a mobile app with 50,000+ registered users.
- Optimized 8 slow PostgreSQL queries using indexing and query rewriting, reducing average response time from 1.2s to 180ms (85% improvement).
- Implemented Redis caching for 5 high-traffic endpoints, reducing database read load by 40%.
- Containerized 3 backend microservices with Docker, enabling zero-downtime deployments.
- Wrote 60+ unit and integration tests using pytest, achieving 85% code coverage on owned modules.
- Collaborated with the mobile team to design API contracts; reduced integration bugs by 30%.

Freelance Backend Developer — Remote

January 2023 – May 2023

- Built a task management REST API using FastAPI and PostgreSQL, deployed on DigitalOcean VPS.
- Integrated Stripe payment gateway processing over PKR 500,000 in monthly transactions.
- Implemented JWT authentication with refresh token rotation and role-based access control.

PROJECTS

Vehicle Monitoring System | Python, OpenCV, YOLOv8, Flask, PostgreSQL

github.com/alihassan/vehicle-monitor | Live demo deployed on Render

- Built a real-time vehicle detection pipeline achieving 92% accuracy on a 5,000-image labeled dataset.
- Processed 30 FPS live video feed using YOLOv8; deployed as a Flask web app with live video streaming.
- Stored detection events in PostgreSQL and built a Chart.js dashboard visualizing 7-day traffic patterns.
- Reduced false positives by 18% through custom confidence-threshold tuning and data augmentation.

E-Commerce REST API | Django REST Framework, PostgreSQL, Redis, Docker, GitHub Actions

github.com/alihassan/ecom-api

- Designed full e-commerce backend with product catalog, cart, order management, and payment modules.
- Handled 500+ concurrent simulated users under Locust load testing; P95 latency stayed under 200ms.
- Set up a CI/CD pipeline with GitHub Actions automating tests and Docker image builds on every push.
- Documented all 30+ API endpoints using Swagger/OpenAPI.

CERTIFICATIONS

- AWS Certified Cloud Practitioner — Amazon Web Services, 2023
- Python for Everybody Specialization (5 courses) — University of Michigan via Coursera, 2022
- PostgreSQL: From Beginner to Advanced — Udemy, 2022